Day 3 — Feature Engineering & Cross-Validation

This report summarizes Day 3 work on the UCI Adult (Census Income) dataset, extending the Logistic Regression and Decision Tree models from Day 2 with principled feature engineering and statistically reliable performance estimation.

Task 1: Engineered Features

Feature	Type	MI Score	Why
log_capital_gain	numeric	0.0843	Compresses skewed capital-gain range
edu_hours_interaction	numeric	0.0829	Combined effect: education + hours
higher_ed	binary	0.0317	College-degree flag vs no degree
has_capital_gain	binary	0.0260	Presence/absence of any capital gain
age_group_40-54	category	0.0253	Peak earning years (life-cycle bin)
hours_category_50+	category	0.0230	Overtime work correlates with income
has_capital_loss	binary	0.0102	Fewer people have capital losses

Key finding: log_capital_gain and edu_hours_interaction are the strongest engineered features (MI > 0.08), capturing magnitude of capital gain and the combined effect of education level with work hours — patterns not explicit in raw columns.

Task 2: Pipeline Rebuild

Integrated engineered features into the ColumnTransformer pipeline via FunctionWrapper. Critical fix: engineer_features() now preserves original 14 columns + adds 7 engineered columns (21 total), enabling the ColumnTransformer to reference column names such as 'age', 'capital_gain', and the new 'age_group', 'hours_category'. Also fixed OneHotEncoder sparse_output=True → False to ensure HistGradientBoostingClassifier compatibility (it requires dense numpy arrays, not scipy sparse matrices).

Task 3: Cross-Validated Model Comparison

5-fold Stratified Cross-Validation with identical preprocessing across three models:

Model	Precision	Recall	F1	ROC AUC
HistGradientBoosting	0.7758 ± 0.0073	0.6526 ± 0.0102	0.7089 ± 0.0078	0.9255 ± 0.0025
Logistic Regression	0.7455 ± 0.0083	0.6184 ± 0.0108	0.6760 ± 0.0093	0.9130 ± 0.0045
Random Forest	0.7274 ± 0.0091	0.6168 ± 0.0072	0.6675 ± 0.0046	0.9021 ± 0.0043

Result: HistGradientBoosting outperforms LR and Random Forest on all four metrics and is the most stable (lowest ROC AUC std = 0.0025). Raw fold precisions: HGB [0.7732, 0.7845, 0.7642, 0.7828, 0.7746] (range 0.020) vs LR [0.7310, 0.7559, 0.7459, 0.7443, 0.7505] (range 0.025).

Task 4: Statistical Comparison & Feature Importance

Metric	Value
Wilcoxon statistic	0.0

p-value	0.0625
HGB won all 5 folds?	YES
Mean precision diff (HGB–LR)	+0.0303 (4.1% relative)
Statistical?	Not significant at α=0.05, but direction unanimous
Practical?	+3pp precision improvement is meaningful for business goal

Feature importance: Tree-based (HGB): top features are edu_hours_interaction, log_capital_gain, age_group_40-54. Linear (LR): log_capital_gain (+5.37), capital_gain (+1.53), Married-civ-spouse (+1.25). Notably, has_capital_gain carries a large negative coefficient (–5.09) due to collinearity with log_capital_gain — confirming redundancy and supporting its removal in feature selection.

Task 5: Feature Selection / Dimensionality Check

Tested SelectKBest(mutual_info_classif) on Logistic Regression: All (122) → Top-50 → Top-30 → Top-15. HGB All vs Top-30 precision: 0.7758 → 0.7647 (drop of 0.0112, 1.4% relative — acceptable). Top-30 selected features by MI: edu_hours_interaction (0.0827), log_capital_gain (0.0801), capital_gain (0.0837), higher_ed (0.0335), age_group_40-54 (0.0251). Engineered features retained in selection: edu_hours_interaction, log_capital_gain, higher_ed, age_group_40-54. Features recommended for drop: has_capital_gain (overlaps log_capital_gain), fnlwgt (sampling weight), low-importance native-country bins.

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Overview of Work Completed

■ Created 7 engineered features from raw columns; MI scores identify strongest signals

■ Rebuilt preprocessing pipeline with FunctionWrapper; fixed column-drop and sparse bugs

■ 5-fold Stratified CV compared LR, RF, HGB — HGB wins (precision 0.7758, ROC AUC 0.9255)

■ Wilcoxon test: p=0.0625 (n=5 power limit), HGB won all 5 folds (unanimous direction)

■ Feature importance: tree-based and LR coefficients confirm engineered features matter

■ Feature selection: Top-30 retains 96% of All-feature precision for HGB (drop only 0.0112)

■ Recommended keep: log_capital_gain, edu_hours_interaction, higher_ed, age_group_40-54, capital_gain, education_num, hours_per_week, marital status, occupation

■ Recommended drop: has_capital_gain (redundant), fnlwgt (non-predictive), weak native-country bins

— End of Day 3 Report —